Section List for preparation in File

a. Theory Question

b. MCQ Question

c. Coding Practice Query

Theory Question : try to cover : 8:00AM to 11:00AM

SQL Interview Questions

❖ DDL (Data Definition Language)

1. What is DDL in SQL?
2. What is the difference between `CREATE`, `ALTER`, and `DROP`?
3. Write a query to create a table with primary key and NOT NULL constraint.
4. How do you add a new column to an existing table?
5. How do you rename a table?
6. What is the difference between `TRUNCATE` and `DELETE`?
7. Can `TRUNCATE` be rolled back?
8. What is a constraint? List types of constraints.
9. What is the difference between `PRIMARY KEY` and `UNIQUE KEY`?
10. How do you remove a column from a table?

❖ DML (Data Manipulation Language)

11. What is DML?
12. What is the difference between `INSERT`, `UPDATE`, and `DELETE`?
13. Write a query to insert multiple records into a table.
14. How do you update multiple columns in a single query?
15. What happens if you run an `update` without a WHERE clause?
16. How do you delete specific records from a table?
17. What is the difference between `DELETE` and `TRUNCATE`?
18. How do you copy data from one table to another?
19. Write a query to insert data from one table into another table.
20. What is a MERGE statement?

❖ TCL (Transaction Control Language)

21. What is TCL?
22. What is a transaction in SQL?
23. What is `COMMIT`?
24. What is `ROLLBACK`?
25. What is `SAVEPOINT`?
26. What happens if a transaction is not committed?
27. Can we rollback after commit?
28. What is auto-commit mode?
29. Difference between `COMMIT` and `ROLLBACK`?
30. What is transaction consistency?

❖ DCL (Data Control Language)

31. What is DCL?
32. What is `GRANT`?
33. What is `REVOKE`?
34. How do you give SELECT permission to a user?
35. How do you remove access from a user?

❖ SELECT Statement

36. What is a SELECT statement?
37. What is the difference between `SELECT *` and selecting specific columns?
38. What is an alias in SQL?
39. How do you remove duplicate records using `SELECT`?
40. What is the DISTINCT keyword?

❖ WHERE Clause

41. What is the WHERE clause?
42. Difference between `WHERE` and `HAVING`?
43. What operators can be used in `WHERE`?
44. What is the BETWEEN operator?
45. What is the IN operator?

❖ GROUP BY

46. What is `GROUP BY`?
47. Why is `GROUP BY` used?
48. What happens if you use `GROUP BY` without aggregate function?
49. Can we use `WHERE` with `GROUP BY`?
50. What are aggregate functions?

❖ HAVING Clause

51. What is the HAVING clause?
52. Difference between `WHERE` and `HAVING`?

53. Can we use `HAVING` without `GROUP BY`?
54. Which executes first: `WHERE` or `HAVING`?
55. Why is `HAVING` needed?

❖ ORDER BY

56. What is `ORDER BY`?
57. What is the default sorting order?
58. Difference between `ASC` and `DESC`?
59. Can we sort by multiple columns?
60. Can we use column aliases in `ORDER BY`?

❖ LIKE Operator

61. What is `LIKE` operator?
62. What are wildcards in SQL?
63. Difference between `%` and `_` in `LIKE`?
64. Write a query to find names starting with 'A'.
65. Write a query to find names ending with 'n'.

❖ Scenario-Based / Practical Questions

66. Find the second highest salary from a table.
67. Find duplicate records in a table.
68. Delete duplicate rows but keep one record.
69. Find employees whose salary is greater than average salary.
70. Count the number of employees in each department.

❖ IN Operator

1. What is the `IN` operator in SQL?
2. How is `IN` different from multiple `OR` conditions?
3. Can `IN` be used with subqueries?
4. What happens if the `IN` list contains `NULL`?
5. Write a query to fetch records using `IN` with multiple values.

❖ BETWEEN Operator

6. What is the `BETWEEN` operator?
7. Is `BETWEEN` inclusive or exclusive?
8. Can `BETWEEN` be used with dates?
9. Difference between `BETWEEN` and using `>=` and `<=`?
10. Write a query using `BETWEEN` for salary range.

❖ ANY Operator

11. What is the `ANY` operator?
12. Difference between `ANY` and `ALL`?
13. How does `ANY` work with comparison operators?
14. Can `ANY` be used without subquery?
15. Write a query using `ANY` with a subquery.

❖ EXISTS Operator

16. What is the `EXISTS` operator?
17. Difference between `EXISTS` and `IN`?
18. When should you use `EXISTS` instead of `IN`?
19. What does `EXISTS` return?
20. Write a query using `EXISTS`.

❖ JOINS

21. What are joins in SQL?
22. Types of joins in SQL?
23. Difference between `INNER JOIN` and `LEFT JOIN`?
24. What is `RIGHT JOIN`?
25. What is `FULL OUTER JOIN`?
26. What is a `CROSS JOIN`?
27. What is a `SELF JOIN`?
28. What happens if no join condition is specified?
29. Difference between `JOIN` and `UNION`?
30. Write a query to join two tables.

MCQ Question (on SQL)

Q1.Which SQL command is used to create a new table?
- A. CREATE TABLE
- B. ALTER TABLE
- C. INSERT INTO
- D. DROP TABLE

Q2.How do you remove a column from an existing table?
- A. DROP COLUMN
- B. REMOVE COLUMN
- C. DELETE COLUMN
- D. ALTER TABLE

Q3.Which command deletes a table completely?
- A. DROP TABLE
- B. DELETE TABLE
- C. TRUNCATE TABLE
- D. REMOVE TABLE

Q4.Which DDL command is used to change the data type of a column?

A. UPDATE COLUMN
B. ALTER TABLE
C. MODIFY TABLE
D. CHANGE COLUMN

Q5.Adding a new column to a table is done with:
A. ADD COLUMN
B. UPDATE TABLE
C. INSERT COLUMN
D. CREATE COLUMN

Q6.Which command renames a table?
A. ALTER TABLE … RENAME TO
B. RENAME TABLE
C. CHANGE TABLE NAME
D. MODIFY TABLE

Q7.Which DDL command is not reversible?
A. CREATE TABLE
B. DROP TABLE
C. ALTER TABLE
D. SELECT

Q8.Creating a table with a primary key column:
A. CREATE TABLE Students(ID INT PRIMARY KEY, Name VARCHAR(50))
B. ALTER TABLE Students ADD ID PRIMARY KEY
C. UPDATE Students SET ID PRIMARY KEY
D. INSERT INTO Students(ID) PRIMARY KEY

Q9.Which DDL statement removes all rows but keeps the structure?
A. TRUNCATE TABLE
B. DELETE
C. DROP TABLE
D. REMOVE

Q10.Which of the following is not DDL?
A. CREATE
B. ALTER
C. INSERT
D. DROP

Q11.Which SQL command is used to insert a new row into a table?
A. INSERT INTO
B. UPDATE
C. DELETE
D. CREATE

Q12.Updating a column value in a table uses which command?
A. INSERT
B. UPDATE

C. ALTER
D. DROP

Q13.To remove rows matching a condition:

A. DELETE FROM TableName WHERE condition
B. DROP TableName
C. TRUNCATE TABLE TableName
D. REMOVE ROW

Q14.Selecting all columns from a table:

A. SELECT * FROM TableName
B. SHOW * FROM TableName
C. DISPLAY * FROM TableName
D. GET * FROM TableName

Q14.Which DML command modifies table data only but not structure?

A. INSERT
B. UPDATE
C. DELETE
D. All of the above

Q15.Insert multiple rows in a single statement:

A. INSERT INTO TableName VALUES (…), (…)
B. INSERT MULTIPLE TableName
C. UPDATE TableName
D. ADD ROWS

Q16. Which command updates multiple columns at once?

A. UPDATE TableName SET Col1 = val1, Col2 = val2 WHERE …
B. ALTER TABLE
C. INSERT INTO
D. DELETE

Q17.DML statements can be rolled back using which command?

A. COMMIT
B. ROLLBACK
C. SAVEPOINT
D. All of the above

Q18. Omitting WHERE in UPDATE:

A. Updates first row
B. Updates all rows
C. Error occurs
D. Modifies table structure

Q19.Which DML command removes all records but keeps table?
A. DELETE
B. TRUNCATE
C. DROP
D. ALTER

Q20.Give SELECT privilege to user Alice:

A. GRANT SELECT ON TableName TO Alice
B. REVOKE SELECT ON TableName FROM Alice
C. COMMIT
D. ROLLBACK

Q21.Revoke INSERT privilege from user Bob:
A. REVOKE INSERT ON TableName FROM Bob
B. GRANT INSERT ON TableName TO Bob
C. DELETE FROM TableName
D. ALTER TABLE

Q23.Which DCL statement is used to give privileges?
A. GRANT
B. REVOKE
C. COMMIT
D. ROLLBACK

Q24. Which command removes all privileges previously granted?
A. REVOKE
B. GRANT
C. DELETE
D. TRUNCATE

Q25. Grant all privileges on a database to a user:
A. GRANT ALL PRIVILEGES ON DatabaseName TO UserName
B. REVOKE ALL PRIVILEGES
C. ALTER DATABASE
D. UPDATE DATABASE

Q26. DCL commands are:
A. COMMIT / ROLLBACK
B. GRANT / REVOKE
C. INSERT / UPDATE
D. CREATE / DROP

Q27. Grant privileges on specific columns only:
A. GRANT SELECT (Col1, Col2) ON TableName TO UserName
B. REVOKE

C. COMMIT
D. ALTER TABLE

Q28. Which is not part of DCL?
A. GRANT
B. REVOKE
C. INSERT
D. Both C & D

Q29. DCL mainly controls:
A. Who can access data
B. Table structure
C. Data values
D. Transactions

Q30. To give execute permission on a procedure:
A. GRANT EXECUTE ON ProcedureName TO UserName
B. REVOKE EXECUTE
C. COMMIT
D. ROLLBACK

Q31. Commit current transaction:
A. COMMIT
B. ROLLBACK
C. SAVEPOINT
D. GRANT

Q32. Undo changes made in a transaction:
A. COMMIT
B. ROLLBACK
C. SAVEPOINT
D. GRANT

Q33. Create a savepoint in a transaction:
A. SAVEPOINT Save1
B. COMMIT
C. ROLLBACK
D. GRANT

Q34. Rollback to a savepoint:
A. ROLLBACK TO Save1
B. COMMIT
C. ALTER TABLE
D. DELETE

Q35. Which TCL command makes changes permanent?
A. COMMIT
B. ROLLBACK
C. SAVEPOINT
D. GRANT

Q36. Which TCL command undoes changes?
A. COMMIT
B. ROLLBACK
C. ALTER TABLE
D. GRANT

Q37. Partial rollback uses:
A. SAVEPOINT
B. COMMIT
C. DELETE
D. DROP

Q38. Auto-commit ON means:
A. Each DML statement is automatically committed
B. Changes need manual COMMIT
C. Table structure modified
D. None

Q39. Which TCL command can rollback partially?
A. ROLLBACK TO Savepoint
B. COMMIT
C. DROP
D. GRANT

Q40. Which TCL command is optional if auto-commit is OFF?
A. COMMIT
B. ROLLBACK
C. Both
D. None

Q41. Select rows where Salary = 30000:
A. WHERE Salary = 30000
B. WHERE Salary > 30000
C. WHERE Salary >= 30000
D. WHERE Salary <> 30000

Q42. Select employees in departments 1, 2, 3:
A. WHERE DeptID IN (1,2,3)
B. WHERE DeptID = 1 AND 2 AND 3

C. WHERE DeptID BETWEEN 1 AND 3
D. WHERE DeptID LIKE '1,2,3'

Q43. Select salaries between 20000 and 50000:
A. WHERE Salary BETWEEN 20000 AND 50000
B. WHERE Salary IN (20000,50000)
C. WHERE Salary >= 20000 AND <= 50000
D. WHERE Salary LIKE '20000-50000'

Q44. Names starting with 'J':
A. LIKE 'J%'
B. LIKE '%J'
C. LIKE '%J%'
D. LIKE '_J%'

Q45. Names ending with 'n':
A. LIKE '%n'
B. LIKE 'n%'
C. LIKE '_n%'
D. LIKE '%_n'

Q46. Names containing 'ar':
A. LIKE '%ar%'
B. LIKE 'ar%'
C. LIKE '%ar'
D. LIKE 'ar'

Q47. Exclude departments 4,5:
A. WHERE DeptID NOT IN (4,5)
B. WHERE DeptID NOT BETWEEN 4 AND 5
C. WHERE DeptID != 4 AND !=5
D. WHERE DeptID <> 4,5

Q48. BETWEEN is:
A. Inclusive
B. Exclusive
C. Only lower inclusive
D. Only upper inclusive

Q49. _ in LIKE matches:
A. Any character
B. Exactly one character
C. No character
D. Only digits

Q50. Combine multiple conditions:

A. WHERE DeptID = 1 AND Salary > 30000
B. WHERE DeptID = 1 OR Salary > 30000
C. WHERE DeptID IN (1,2) AND Salary BETWEEN 20000 AND 50000
D. All of the above

MCQ Question on Aptitude

Aptitude Question (percentage, average, ratio and proportion)

PART 1: Percentage (10 Questions)

Q1. What is 20% of 250?

A. 40
B. 50
C. 60
D. 70

Answer: B

Solution:
20% means 20/100. So, 20/100 × 250 = 50.
You can also think of 10% of 250 = 25, so 20% = 25 × 2 = 50.
Hence, the correct answer is 50.

Q2. A number increases from 100 to 120. What is the percentage increase?

A. 10%
B. 15%
C. 20%
D. 25%

Answer: C

Solution:
Increase = 120 − 100 = 20.

Percentage increase = (20/100) × 100 = 20%.
Always divide increase by original value.

Q3. Find 15% of 80.

A. 10
B. 12
C. 14
D. 16

Answer: B

Solution:
15% = 15/100.
So, 15/100 × 80 = 12.
Alternatively, 10% of 80 = 8 and 5% = 4 → total = 12.

Q4. A number is decreased by 25% and becomes 75. What was the original number?

A. 80
B. 90
C. 100
D. 120

Answer: C

Solution:
After 25% decrease, value becomes 75% of original.
So, 75% = 75 → original = 75 / 0.75 = 100.
Thus original number is 100.

Q5. What is 50% of 300?

A. 100
B. 120
C. 150
D. 200

Answer: C

Solution:
50% means half.
Half of 300 = 150.
So, directly answer is 150.

Q6. A student scored 45 out of 60. Find percentage.

A. 70%
B. 75%
C. 80%
D. 85%

Answer: B

Solution:
Percentage = (45/60) × 100.
= 75%.
Divide first → 45 ÷ 60 = 0.75.

Q7. What is 10% of 450?

A. 40
B. 45
C. 50
D. 55

Answer: B

Solution:
10% means divide by 10.
450 ÷ 10 = 45.
So answer is 45.

Q8. Price increased from 200 to 260. % increase?

A. 20%
B. 25%
C. 30%
D. 35%

Answer: C

Solution:
Increase = 260 − 200 = 60.
% increase = (60/200) × 100 = 30%.
Always compare with original.

Q9. 40% of a number is 200. Find number.

A. 400
B. 450
C. 500
D. 600

Answer: C

Solution:
40% = 200.
Number = 200 / 0.4 = 500.
So original number is 500.

Q10. What is 5% of 600?

A. 20
B. 25
C. 30
D. 35

Answer: C

Solution:
5% = 5/100.
600 × 5/100 = 30.
Also 10% is 60, so half = 30.

Q11. Average of 2, 4, 6, 8?

A. 4
B. 5
C. 6
D. 7

Answer: B

Solution:
Sum = 2+4+6+8 = 20.
Average = 20/4 = 5.
Average = total / number of values.

Q12. Average of 5 numbers is 20. Total sum?

A. 80
B. 90
C. 100
D. 110

Answer: C

Solution:
Average × number = total.
20 × 5 = 100.
So total sum is 100.

Q13. Average of 10 numbers is 15. Find sum.

A. 120
B. 130
C. 150
D. 160

Answer: C

Solution:
Sum = 15 × 10 = 150.
Direct formula: total = average × count.

Q14. Average of 3 numbers is 30. Find total.

A. 60
B. 70
C. 80
D. 90

Answer: D

Solution:
Total = 30 × 3 = 90.
Always multiply average by count.

Q15. Average of 20 and 40?

A. 20
B. 25
C. 30
D. 35

Answer: C

Solution:
(20 + 40)/2 = 60/2 = 30.
Average of two numbers is midpoint.

Q16. If average is 50 for 4 numbers, sum?

A. 150
B. 200
C. 250
D. 300

Answer: B

Solution:
Sum = 50 × 4 = 200.
Formula: sum = average × count.

Q17. Average of 6 numbers is 10. Add 1 number (20). New average?

A. 11
B. 12
C. 13
D. 14

Answer: B

Solution:
Old sum = 6 × 10 = 60.
New sum = 60 + 20 = 80.
New average = 80/7 ≈ 11.43 ≈ 12 (closest).

Q18. Average of 3, 6, 9?

A. 5
B. 6
C. 7
D. 8

Answer: B

Solution:
Sum = 18.
Average = 18/3 = 6.
Simple arithmetic mean.

Q19. Average of 4 numbers is 25. One number is 10. Remaining sum?

A. 70
B. 80
C. 90
D. 100

Answer: C

Solution:
Total sum = 25 × 4 = 100.
Remaining = 100 − 10 = 90.

Q20. Average of 1 to 9?

A. 4
B. 5
C. 6
D. 7

Answer: B

Solution:
For consecutive numbers: average = (first + last)/2.
(1 + 9)/2 = 5.
Shortcut formula.

Q21. Ratio of 10:20 simplifies to?

A. 1:2
B. 2:3
C. 3:4
D. 1:3

Answer: A

Solution:
Divide both by 10 → 10:20 = 1:2.
Simplify ratios like fractions.

Q22. If a:b = 2:3 and b:c = 4:5, find a:c.

A. 2:5
B. 4:5
C. 8:15
D. 10:15

Answer: C

Solution:
Make b same:
2:3 and 4:5 → multiply first by 4 → 8:12
Second by 3 → 12:15
So a:c = 8:15.

Q23. Divide 100 in ratio 2:3.

A. 40, 60
B. 50, 50
C. 30, 70
D. 20, 80

Answer: A

Solution:
Total parts = 5.
First = $(2/5)\times100 = 40$
Second = $(3/5)\times100 = 60$.

Q24. Ratio 3:4, if first is 30, second?

A. 35
B. 40
C. 45
D. 50

Answer: B

Solution:
$3 \rightarrow 30$ means multiply by 10.
So $4 \times 10 = 40$.

Q25. If x:y = 5:2 and x=50, find y.

A. 10
B. 20
C. 25
D. 30

Answer: B

Solution:
$5 \rightarrow 50$ means multiply by 10.
So $2 \times 10 = 20$.

Q26. Ratio of 15 and 45?

A. 1:2
B. 1:3
C. 2:3
D. 3:5

Answer: B

Solution:
Divide both by 15 → 1:3.

Q27. If a:b = 4:5, total = 90. Find a.

A. 30
B. 40
C. 50
D. 60

Answer: B

Solution:
Total parts = 9.
$a = (4/9) \times 90 = 40$.

Q28. If ratio is 7:3, difference = 40. Find numbers.

A. 70, 30
B. 60, 20
C. 80, 40
D. 90, 50

Answer: A

Solution:
Difference parts = 4.
4 parts = 40 → 1 part = 10.
Numbers = 70 and 30.

Q29. If a:b = 1:2, and b:c = 3:4, find a:c.

A. 1:4
B. 3:4
C. 3:8
D. 1:8

Answer: C

Solution:
Match b:
1:2 and 3:4 → multiply first by 3 → 3:6
Second by 2 → 6:8
So a:c = 3:8.

Q30. Divide 60 in ratio 1:2:3.

A. 10, 20, 30
B. 15, 20, 25
C. 5, 10, 15
D. 20, 20, 20

Answer: A

Solution:
Total parts = 6.
Values: $(1/6)\times 60=10$, $(2/6)=20$, $(3/6)=30$.
Hence result is 10, 20, 30.

Practice Query

◆ SELECT Statement

1. Retrieve the top 3 highest-paid employees from a table without using `LIMIT` or `TOP`.
2. Write a query to display employee names concatenated with their department name, with the alias `EmployeeDept`.
3. Select all records where the concatenation of first and last name contains more than 12 characters.
4. Retrieve only unique combinations of department and job title from a table.
5. Write a query to show the total length of all employee names in the table.

◆ WHERE Clause

6. Find employees whose names contain 'a' but do not start or end with 'a'.
7. Retrieve records where the salary is above the average for employees in the same department.
8. Write a query to find employees who joined in the last 6 months and earn more than 50,000.
9. Find employees whose ID is not in a given set of values but exists in another table.
10. Write a query to find employees with salaries between the second and third quartile of the salary range.

◆ GROUP BY & Aggregate Functions

11. Show the department with the second highest total salary.
12. Count the number of employees in each department who earn above the overall average salary.
13. Retrieve departments with more than 5 employees, but show only those with average salary below 70,000.
14. List each department and the maximum, minimum, and average salary in a single row.
15. Group employees by their hire year and month, and show the count per month.

◆ HAVING Clause

16. Retrieve departments where the total salary exceeds 1,000,000 but exclude departments with only one employee.
17. Write a query to find job titles where the average salary is higher than the overall company average.
18. Find departments where all employees earn more than 40,000.
19. Retrieve the number of employees in departments that have more than 10 employees earning above 70,000.
20. Show the departments with exactly 3 employees having the same job title.

◆ ORDER BY

21. Sort employees first by department ascending, then by salary descending, then by hire date ascending.
22. Write a query to show the top 5 salaries per department.
23. Order employees by the length of their full name.
24. Sort employees by the number of vowels in their names, descending.
25. Display records where salaries are sorted in descending order but with NULL salaries at the top.

❖ LIKE Operator

26. Find all employees whose name has two consecutive vowels.
27. Retrieve names where the second character is 'a' and ends with 'n'.
28. Find names that contain either 's' or 't' but not both.
29. Write a query to find names that start and end with the same letter.
30. Retrieve names where the third character is a number.

❖ IN Operator

31. Find employees who work in departments that have fewer than 10 employees using `IN`.
32. Write a query to find salaries that exist in a predefined set and are above the median salary.
33. Retrieve employees whose manager ID is in a subquery selecting all managers who have more than 3 direct reports.
34. Find records where the department ID is in a dynamic list returned by another table.
35. Write a query to exclude employees whose salaries appear in the top 5 salaries using `NOT IN`.

❖ BETWEEN Operator

36. Find employees hired between two specific dates, excluding the boundary dates.
37. Retrieve salaries within the interquartile range (25th–75th percentile).
38. Find records with salaries not between 30,000 and 60,000.
39. Write a query to find employees whose salaries fall in overlapping ranges of two departments.
40. Retrieve employees whose hire dates are between the earliest and latest hire dates in the table.

❖ ANY / ALL Operator

41. Find employees whose salary is greater than any employee in the 'IT' department.
42. Retrieve employees whose salary is less than all salaries in a subquery selecting executives.
43. Find employees whose salary is greater than any salary in a list of high performers but less than the maximum salary overall.
44. Write a query to compare salaries across departments using `ALL`.
45. Retrieve employees whose salary matches at least one of the top 3 salaries across all departments.

❖ EXISTS Operator

46. Retrieve employees who exist in departments that have more than 10 employees.
47. Find employees who exist in a table of projects assigned to at least one active project.
48. Write a query to select employees only if their manager exists in the same table.
49. Find records in one table where there is no corresponding record in another table using `EXISTS`.
50. Retrieve departments that have employees with salaries above a certain threshold using `EXISTS`.

❖ JOINS

51. Write a query to list all employees along with their manager's name using self-join.
52. Retrieve employees and departments including departments that have no employees.
53. Find employees working in multiple departments using a join.
54. Perform a cross join between employees and projects and filter by salary > 50,000.
55. List employees along with projects, but only for employees who have no projects (use join cleverly).
56. Write a query to combine two tables of transactions and invoices without using `UNION`.
57. Find duplicate records across two tables using a join.
58. Perform a full outer join between two tables and highlight missing values from each side.
59. Retrieve employees whose department name starts with 'S' using a join with a department table.
60. Write a query that ranks employees by salary in each department using joins and aggregate functions.